Silver Point Gateway - Peecho API Technical Specification

Gateway Provider: Silver Point Print-on-Demand • **Partner:** Peecho B.V. (Netherlands) • **Protocol:** REST JSON over HTTPS (TLS 1.3)

1. Environment Endpoints

Environment	Method & Full Webhook URL	Secret Key Variable	Order Flagging
Production	`POST https://orderflow.silverpointprint.com/api/v1/peecho/webhook/`	`PEECHO_PRODUCTION_SECRET_KEY`	`is_sandbox = False` (Live)
Staging Sandbox	`POST https://staging.orderflow.silverpointprint.com/api/v1/peecho/webhook/`	`PEECHO_STAGING_SECRET_KEY`	`is_sandbox = True` (TEST_ORDER)

2. Secret Key & HMAC-SHA256 Authentication

- **Signature Calculation:** Every inbound webhook request must include the `X-Peecho-Signature` HTTP header computed as the hex-encoded HMAC-SHA256 digest of the raw request body using the environment's secret key.
- **Isolated Environment Keys:** Production requests on `orderflow.silverpointprint.com` are verified against `PEECHO_PRODUCTION_SECRET_KEY`. Staging requests on `staging.orderflow.silverpointprint.com` are verified against `PEECHO_STAGING_SECRET_KEY`.
- **Secure Environment Storage:** Both secret keys are stored strictly in the server's `.env` configuration and verified on every inbound request using constant-time digest comparison (`hmac.compare_digest`).

```
# Required Inbound HTTP Headers
Content-Type: application/json
X-Peecho-Signature: sha256=8f4e2b91c0d4a7e3190f8c21d6b5e4a9012c3d4e5f6a7b8c9d0e1f2a3b4c5d6e

# Environment Configuration (.env)
PEECHO_PRODUCTION_WEBHOOK_URL=https://orderflow.silverpointprint.com/api/v1/peecho/webhook/
PEECHO_STAGING_WEBHOOK_URL=https://staging.orderflow.silverpointprint.com/api/v1/peecho/webhook/
PEECHO_PRODUCTION_SECRET_KEY=sp_live_<64_char_hex_secret>
PEECHO_STAGING_SECRET_KEY=sp_stg_<64_char_hex_secret>
```

3. Inbound API (Peecho → Silver Point)

Endpoint: `POST /api/v1/peecho/webhook/` • **Purpose:** Ingests order metadata, customer delivery address, paper options, and print PDF links. Requests arriving via the Staging domain or authenticated with `PEECHO_STAGING_SECRET_KEY` are automatically saved with `is_sandbox = True`.

```
{
  "event": "order.created",
  "order_id": "PEECHO-904120",
  "priority": "EXPRESS_24H",
  "currency": "EUR",
  "total_amount": "48.50",
  "carrier": "DHL Express",
  "customer": { "name": "Jan de Vries", "email": "jan@example.nl", "phone": "+31 20 794 0100" },
  "shipping_address": {
    "recipient_name": "Jan de Vries",
    "street": "Keizersgracht 241",
    "city": "Amsterdam",
    "postal_code": "1016 EA",
    "country": "NL"
  },
  "items": [
    {
      "item_id": "ITEM-01",
      "product_type": "Hardcover Photo Book A4",
      "quantity": 1,
      "page_count": 64,
      "spine_width_mm": 9.5,
      "paper_stock": "200gsm Silk Inner + 350gsm Matte Cover",
      "pdf_artwork_url": "https://cdn.peecho.com/print/904120/interior.pdf",
      "pdf_cover_url": "https://cdn.peecho.com/print/904120/cover.pdf"
    }
  ]
}
```

HTTP Status	Response Code	Description & Gateway Action
201 Created	`PROCESSED`	Order validated and created. Returns `order_number`, `environment`, and `is_sandbox`.
200 OK	`DUPLICATE_IGNORED`	Idempotent SHA-256 payload hash match; duplicate order ignored safely.
400 Bad Request	`INVALID_PAYLOAD`	Malformed JSON or missing required fields (`order_id`, `items`).
401 Unauthorized	`INVALID_SIGNATURE`	HMAC-SHA256 digest mismatch in `X-Peecho-Signature`; logged as `INVALID_AUTH`.

4. Outbound APIs (Silver Point → Peecho)

A. Outbound Shipment Callback (`POST /v2/orders/{id}/shipment`)

Triggered automatically when the courier AWB barcode is scanned at Dispatch. Sends the carrier name, tracking code, dispatch timestamp, tracking URL, and AWB document link back to Peecho.

```
{
  "peecho_order_id": "PEECHO-904120",
  "order_reference": "ORD-2026-1001",
  "carrier": "DHL Express",
  "tracking_code": "1234567890",
  "shipment_date": "2026-11-04T14:30:00Z",
  "tracking_url": "https://www.dhl.com/en/express/tracking.html?AWB=1234567890",
  "awb_document_url": "https://orderflow.silverpointprint.com/media/awb/ORD-2026-1001_awb.pdf"
}
```

B. Order Status & Exception Callbacks (`POST /v2/orders/{id}/status`)

Whenever an order transitions across production milestones or encounters an exception (e.g., `ORDER_CANCELLED`, `PRINT_FAILED`, `QC_REJECTED`), Silver Point pushes a real-time status payload to Peecho.

Internal Event	Outbound Peecho Status	Trigger Condition & Action
`PREPRESS_PASSED`	`PREFLIGHT_VERIFIED`	PDF artwork verified for resolution, spine width, and bleed.
`IN_PRODUCTION / PRINTING`	`IN_PRODUCTION`	QR Job Ticket scanned at Press / Bindery station.
`EXCEPTION_HOLD`	`QC_REJECTED / PRINT_FAILED`	Artwork preflight error or physical print defect flagged (auto-requeued for reprint).
`CANCELLED`	`ORDER_CANCELLED`	Order cancelled prior to dispatch; stops physical press execution.
`DELIVERED`	`DELIVERED`	Final delivery confirmation received from carrier tracking poller.

```
{
  "peecho_order_id": "PEECHO-904120",
  "order_reference": "ORD-2026-1001",
  "status": "QC_REJECTED",
  "reason_code": "PRINT_STREAKING",
  "action_taken": "AUTO_REQUEUED_RUSH_8H",
  "notes": "Print defect detected at QC station; replacement job queued automatically.",
  "updated_at": "2026-11-04T11:15:00Z"
}
```

C. Network Error Handling & Exponential Backoff Schedule

If an outbound callback to Peecho returns a `5xx` server error, `429 Too Many Requests`, or network timeout, Silver Point automatically retries according to the backoff schedule below:

Attempt	Backoff Delay	Retry Condition	Fallback / Escalation Action
Attempt 1 (Initial)	Immediate (`0s`)	Real-time scan event at Press / QC / Dispatch	Logs HTTP status; queues Retry #1 on timeout or `5xx`.
Retry #1	**+30 Seconds**	Transient network blip or gateway timeout	Re-sends signed JSON payload with idempotency key.
Retry #2	**+2 Minutes**	Upstream endpoint still unreachable	Increments `peecho_sync_attempts` counter.
Retry #3	**+10 Minutes**	Sustained API maintenance window	Flags warning indicator on POD Integrations dashboard.
Retry #4 (Final)	**+1 Hour**	Extended outage recovery attempt	If failed, marks `SYNC_FAILED` and alerts integration admin for manual replay.